Computer Graphics

Professor YoungWoon Cha

Computer Science and Engineering
Metaverse Convergence
Graduate School

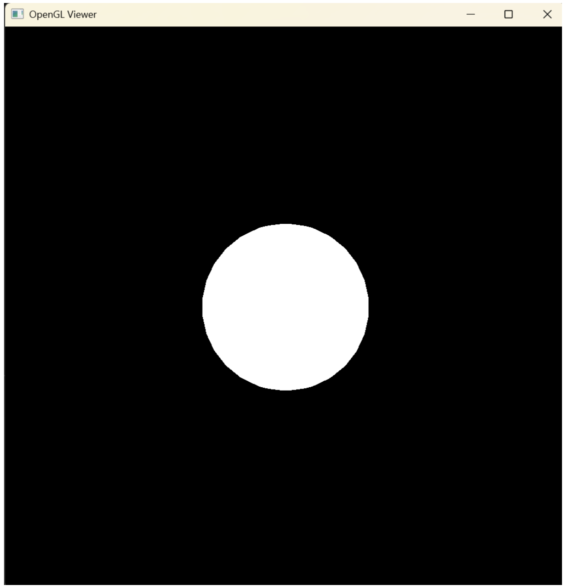
Contents

- Assignment 5
- Questions
- Instructions

Assignment 5

Implementing a Rasterizer

- Objective
  - The goal of this assignment is to write a simple rasterization pipeline and use it to draw a single shaded sphere, which is in turn represented using multiple triangles.
  - These triangles (and their vertices) are generated using code that is provided on the course webpage.
  - For reference, a view of the scene is shown below.

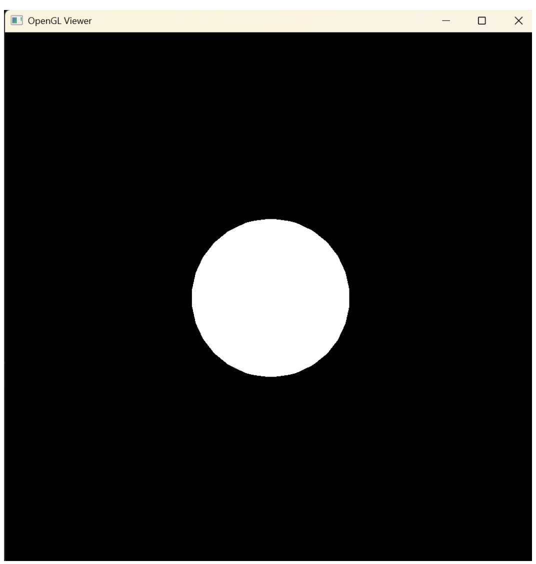


Unshaded Image using the triangle rasterizer

Questions

Q1: Transformations and Rasterization

- Write a simple rasterization pipeline to draw a single shaded sphere.
  - Using the provided code, generate triangles for a unit sphere.
  - Implement a transformation pipeline for rasterization.
  - Implement triangle rasterization.
  - Back-face culling is optional, but you must use a depth buffer to handle occlusion.
  - Display your result as an unshaded image as shown here.



Unshaded Image using a triangle rasterizer

Your output must be identical to the image shown here.

Q1: Transformations and Rasterization

- Use the following sequence of transformations to display the sphere:
  - The provided code, "sphere_scene.cpp", generates triangles for a unit sphere.
  - Use a modeling transform to transform the unit sphere to a sphere centered at $(0, 0, -7)$ with a radius of 2.
  - Set the eye point at $e = (0, 0, 0)$
  - Set the orientation to $u = (1, 0, 0)$, $v = (0, 1, 0)$ and $w = (0, 0, 1)$.
  - (Note that the camera is looking along the direction $-w$.)
  - Use a perspective projection transform with $l = -0.1$, $r = 0.1$, $b = -0.1$, $t = 0.1$, $n = -0.1$, $f = -1000$.
  - Use a viewport transform with $nx = 1024$, and $ny = 1024$.

Q1: Transformations and Rasterization

- In your presentation, you must clearly explain and demonstrate your implementation of the following:
  - Your transformation pipeline.
  - Your triangle rasterization.
  - Your depth buffer to handle occlusion.

Q2: Implement a New Feature

- Implement and demonstrate a custom feature
  - For example:
    - Adding differently shaped objects (other than spheres)
    - Adding multiple objects or moving objects over time
    - Implementing different rasterization techniques, such as line rasterization
    - Using different camera models, such as orthographic projection

  - This question will be graded on a relative scale (A/B/C).
  - Fewer than 50% of students will receive an 'A' grade.

Instructions

Submission Guidelines

- Important Notes
  - Your code must include a separate project for each question but with a single solution.
  - You are not allowed to use OpenGL to perform hardware rasterization.
    - You must implement your own transformation pipeline, rasterizer, and shading algorithms.
  - Along with your code, please submit output screenshots for each question and explain them during your presentation.

Submission Guidelines

- Submission Requirements
  - Presentation video: Under 5 minutes and 30MB maximum (MP4 format)
  - Presentation slides: PDF format
  - Source code + output screenshots: VS 2022 project and image files (ZIP format)

Summary

- Assignment 5
- Questions
- Instructions

End of Class

Thank you

E-mail: youngcha@konkuk.ac.kr